

METZ FRESCATY, France

WMO: 070900

Lat: 49.072N

Lon: 6.132E

Elev: 192

StdP: 99.04

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-5.9	-11.7	1.4	-5.4	-9.4	1.7	-4.0	12.0	9.6	10.9	8.8	1.5	20	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	31.0	20.0	29.2	19.4	27.3	18.7	21.2	28.5	20.4	27.1	19.5	25.6	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.0	23.8	18.0	13.3	22.8	17.2	12.6	22.0	62.3	28.7	59.3	27.2	56.3	25.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.9	-10.6	34.3	3.0	2.7	-12.8	36.2	-14.6	37.8	-16.3	39.3	-18.5	41.2		
(5)			-10.7	22.8	3.3	0.6	-13.1	23.3	-15.0	23.6	-16.9	24.0	-19.3	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	2.3	3.4	6.7	10.2	14.5	17.7	19.4	19.3	15.3	11.1	6.4	2.8
	DBStd	7.16	4.40	4.05	3.73	3.57	3.41	3.45	3.11	3.23	2.94	3.52	3.68	4.36
	HDD10.0	954	239	186	114	40	4	0	0	0	1	29	116	225
	HDD18.3	2911	497	418	361	243	125	54	24	26	99	223	358	482
	CDD10.0	1247	1	1	11	47	145	229	290	289	159	64	8	2
	CDD18.3	164	0	0	0	0	8	34	56	57	8	1	0	0
	CDH23.3	1683	0	0	0	14	109	353	569	549	86	3	0	0
	CDH26.7	529	0	0	0	1	18	101	189	204	16	0	0	0
Wind	WSAvg	2.9	3.3	3.3	3.3	3.0	2.8	2.7	2.9	2.5	2.5	2.6	2.8	3.3
Precipitation	PrecAvg	768	65	56	64	50	71	71	64	65	60	64	68	70
	PrecMax	1045	137	167	155	160	158	151	128	143	146	176	134	172
	PrecMin	475	25	12	6	13	24	0	15	10	14	2	10	4
	PrecStd	140	29	36	31	29	27	36	32	34	33	44	28	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.7	20.3	25.3	29.1	32.1	33.2	36.3	28.8	23.2	16.5	13.3
		MCWB	11.1	9.6	11.8	15.7	19.9	19.8	20.9	20.4	19.0	17.6	12.8	11.2
	2%	DB	11.6	12.5	17.4	22.6	26.1	29.5	30.7	31.1	26.0	20.3	14.3	11.8
		MCWB	9.9	9.0	11.0	14.0	17.7	19.6	19.9	20.3	18.1	15.8	12.0	10.1
	5%	DB	10.0	10.8	14.9	20.0	24.2	27.4	28.9	28.9	23.4	18.4	12.9	10.4
		MCWB	8.5	8.4	10.0	12.7	16.2	19.0	19.3	19.3	16.9	15.0	11.1	8.9
	10%	DB	8.6	9.2	12.9	17.6	22.0	25.1	26.8	26.4	21.3	16.6	11.6	9.0
		MCWB	7.2	7.4	9.5	11.7	15.1	17.8	18.3	18.5	15.8	14.0	10.1	7.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	10.7	13.3	16.4	20.4	22.1	22.2	22.2	20.1	18.1	13.6	11.6
		MCDB	12.6	13.1	17.9	23.7	27.2	29.0	30.1	30.6	25.8	21.5	15.3	12.8
	2%	WB	10.1	9.6	11.9	14.7	18.8	20.7	21.1	21.1	18.9	16.7	12.3	10.2
		MCDB	11.4	11.7	15.4	21.1	24.8	27.4	28.4	28.8	24.2	19.6	14.0	11.4
	5%	WB	8.6	8.7	10.9	13.3	17.1	19.6	20.1	20.2	17.7	15.4	11.3	9.0
		MCDB	9.9	10.5	14.3	18.5	22.0	25.7	26.9	27.0	22.0	18.0	12.8	10.3
	10%	WB	7.2	7.6	9.7	12.2	15.9	18.5	19.1	19.2	16.7	14.2	10.2	7.7
		MCDB	8.5	9.2	12.5	16.8	20.6	24.0	24.8	25.1	20.3	16.2	11.5	8.9
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.7	8.8	10.6	10.6	10.9	10.6	10.7	10.0	8.5	5.6	4.6
		MCDBR	6.2	9.5	13.2	15.4	14.8	14.8	14.6	15.6	14.3	11.4	7.4	5.7
		MCWBR	5.2	6.7	7.8	7.9	7.2	6.5	5.5	6.0	7.1	7.1	5.4	4.8
	5% WB	MCDBR	6.0	8.2	10.6	13.7	13.1	13.2	13.3	13.1	11.8	10.1	6.8	5.6
		MCWBR	5.3	6.2	6.7	7.4	6.8	6.4	5.5	5.6	6.3	6.7	5.3	5.0
Clear-Sky Solar Irradiance	taub	0.312	0.338	0.365	0.396	0.405	0.409	0.405	0.401	0.375	0.359	0.330	0.310	
	taud	2.434	2.384	2.316	2.252	2.258	2.279	2.285	2.323	2.374	2.409	2.431	2.417	
	Ebn at Noon	746	807	841	850	855	852	850	836	822	769	713	693	
	Edh at Noon	64	84	106	125	130	128	126	116	100	82	64	58	
All-Sky Solar Radiation	RadAvg	0.86	1.60	2.81	4.29	5.07	5.71	5.39	4.64	3.53	2.06	0.98	0.68	
	RadStd	0.13	0.29	0.41	0.56	0.56	0.50	0.53	0.49	0.38	0.17	0.10	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page